WINDLENS

Energy intelligence for UK day-ahead power trading

Video Demo Script

A complete customer pitch and technical walkthrough for the GridBricks full-stack challenge.

Target duration: 8 to 9 minutes (maximum 10 minutes)

Live application: https://windlens-two.vercel.app/

Recording goal: Present the product as a useful decision-support pilot, then make the architecture and engineering decisions easy for a technical reviewer to understand.

Recording principle

Show the product first, connect every screen to a trading decision, and be transparent about data provenance. Do not describe synthetic fallback values as live market data.

Run of show

00:00-00:10	**Opening**
00:10-04:50	**Part 1 - Customer presentation**
04:50-09:50	**Part 2 - Technical walkthrough**

Opening | 10 seconds

On screen: Open the deployed WindLens application in a clean browser window.

Say: Hello, and welcome to WindLens - an energy intelligence dashboard designed to support UK day-ahead electricity procurement decisions. This pilot focuses on national wind generation forecasts and forecast accuracy using Elexon BMRS data.

PART 1

Customer presentation | approximately 4 minutes

1. Business problem | 45 seconds

On screen: Show the Decision Brief page.

Say: UK energy utilities need to buy electricity from the wholesale market before the delivery period begins. A key uncertainty is renewable generation, particularly wind generation. If wind output is lower than expected, the utility may need to procure additional electricity. If it is higher than expected, the utility may over-procure and increase balancing exposure. WindLens helps the trading desk understand the latest forecast and how reliable that forecast has been.

2. Decision Brief | 60 seconds

On screen: Point to the delivery date, latest forecast, chart, accuracy KPI, MAE KPI, decision signal, and decision rule.

Say: This is the Decision Brief view. The selected delivery date is shown at the top and the pilot is limited to January 2026. The main chart compares forecast wind generation with actual wind generation across 30-minute delivery intervals. The green line represents the forecast and the dashed line represents actual generation. Forecast accuracy gives a quick summary, while Mean Absolute Error represents the average difference between forecast and actual generation in megawatts. The decision signal highlights the forecast peak and explains how MAE can be used as a procurement buffer. WindLens selects the latest WINDFOR forecast published at or before 16:00 UTC on the day before delivery, then compares it with FUELHH WIND actual generation.

3. Business value | 45 seconds

On screen: Move between the chart, MAE card, and Decision Signal card.

Say: The goal is not simply to display a graph. The goal is to support a better trading decision. If the forecast predicts 1,200 megawatts of wind generation but the average forecast error is 100 megawatts, the desk can account for that uncertainty when deciding how much additional electricity to procure. WindLens does not make the decision automatically; it gives the trader a transparent view of the forecast, the error, and the uncertainty around the renewable position.

4. Forecast Explorer | 60 seconds

On screen: Open Forecast Explorer. Show the chart, filters, table, and Download CSV button.

Say: The Forecast Explorer provides a detailed operational view. The table contains each half-hourly interval, the forecast, the actual value, and the error in megawatts. The trader can filter all intervals, over-forecast intervals, or under-forecast intervals. The CSV export allows the analyst to download matched intervals for further review or internal trading analysis.

5. Accuracy Analysis | 50 seconds

On screen: Open Accuracy Analysis and show Accuracy, MAE, RMSE, MAPE, Bias, confidence, and error distribution.

Say: The Accuracy Analysis page quantifies forecast quality. MAE shows average absolute error. RMSE gives more weight to large errors, which is useful because unusually large misses can create procurement risk. MAPE expresses error as a percentage, while bias shows the direction of the error. Positive bias indicates over-prediction; negative bias indicates under-prediction. The confidence view gives a quick summary and the error distribution shows whether errors are concentrated around zero or skewed.

6. Dark mode and customer close | 30 seconds

On screen: Toggle dark mode.

Say: WindLens also supports a class-based dark mode for analysts who work with the dashboard for long periods. In summary, WindLens gives the trading desk a focused view of the latest forecast, forecast versus actual generation, forecast accuracy, procurement uncertainty, and interval-level errors.

PART 2

Technical walkthrough | approximately 5 minutes

1. Architecture | 45 seconds

On screen: Show the repository in VS Code.

Say: WindLens is implemented as a React frontend with a Python FastAPI backend. The frontend uses React, Vite, and Tailwind CSS utilities. The backend retrieves and processes Elexon BMRS data. The frontend consumes structured API responses and renders the Decision Brief, Forecast Explorer, Accuracy Analysis, and Monthly Trends views. The frontend is deployable to Vercel and the backend is containerized with Docker for a service such as Render.

2. Frontend structure | 60 seconds

On screen: Show frontend/src: App.jsx, components, hooks, pages, utils, and config.js.

Say: The frontend is organized by responsibility. App.jsx owns application-level state such as active page, selected date, and dark mode. Shared configuration is in config.js. Reusable components include the sidebar, header, date picker, chart, and metric cards. Page-specific views are separated into Overview, Forecast Explorer, Accuracy, and Monthly Trends. The useForecastData hook owns API requests, loading state, errors, and derived chart data. Date formatting and forecast calculations are isolated in utility modules.

3. Backend and Elexon processing | 75 seconds

On screen: Open backend/main.py or the FastAPI docs.

Say: The main endpoint is GET /api/forecast?delivery_date=2026-01-15. FUELHH provides actual generation values and is filtered to fuelType WIND. startTime identifies each 30-minute delivery interval. WINDFOR provides forecasts with startTime, publishTime, and generation. For each interval, WindLens selects the newest forecast published at or before the 16:00 UTC day-ahead cutoff. This prevents using information that would not have been available when the purchasing decision was made. The backend joins forecasts with actual WIND generation and calculates Accuracy, MAE, RMSE, MAPE, and Bias.

4. API and data transparency | 45 seconds

On screen: Show the API response or /docs.

Say: The API response contains the delivery date, source, decision cutoff, dataset names, publish times, forecast and actual values, and calculated metrics. The frontend does not contain business KPI values as hard-coded constants. It derives the displayed metrics from the API response. The application distinguishes live Elexon data, unavailable data, and local demonstration fallback data. If matching historical data is unavailable, the UI shows an explicit error instead of presenting synthetic numbers as live market data.

5. Docker and deployment | 50 seconds

On screen: Show backend/Dockerfile, frontend/Dockerfile, docker-compose.yml, and render.yaml.

Say: Docker makes the backend runtime reproducible. The backend image installs FastAPI dependencies and runs Uvicorn on port 8000. The frontend image builds the Vite application and serves production files on port 4173. Docker Compose starts both services locally. The frontend can be deployed to Vercel and the backend as a Docker service using Render. VITE_API_URL configures the frontend endpoint, while CORS_ORIGINS configures permitted frontend origins.

6. Development practices and close | 45 seconds

On screen: Show the terminal running npm run lint, npm run build, and the backend syntax/smoke check.

Say: The project includes frontend linting and production build checks, backend syntax validation, API smoke tests, explicit loading and error states, responsive design, dark mode, data provenance, and January 2026 scope enforcement. The next production step would be connecting the application to an archived January 2026 Elexon export or historical endpoint so the deployed environment can run fully from verified historical observations. This completes the technical walkthrough of WindLens.

Final recording checklist

Open the Vercel URL in an incognito window. Test Decision Brief, Forecast Explorer, Accuracy Analysis, Monthly Trends, date navigation, dark mode, CSV export, and the API error state. Keep the final video below 10 minutes. Upload to YouTube as Unlisted and test the unlisted link while signed out. Submit the GitHub URL, Vercel URL, backend health URL, and YouTube URL in the company form.

Application URL: https://windlens-two.vercel.app/